

Thuhina Water – Technology Stack, Hosting & Communication

Product: Water Distribution Inventory & Sales Management System **Based on:** NUVI IT – Technology Stack & Communication Module (Unified Technical Reference, Aug 2026) **Related:** BRD v4.0 (BRD Water Distribution System.md) · System Architecture & Design v1.1 · Clickable prototype (/prototype) **Version:** 1.1 · **Date:** 08/10/2026 (v1.0: 04/10/2026) **Changes in 1.1:** aligned with BRD v4.0 – new printed documents (customer quotations, dispatch notes, expense statements), quotations emailed from the user's own email program, sizing note for agreed prices and expenses. The stack, tier, hosting and costs are unchanged.

1. Purpose & Scope

This document applies the shared NUVI IT technology stack to the Thuhina Water system. It sets out the deployment tier, hosting, backup, communication (email / WhatsApp) and running costs for this customer.

Sizing inputs from the BRD

Factor	Thuhina Water
System users	~5–8 (Admin ×2, Accountant ×1, Delivery Staff ×3–5)
Concurrent users	1–3 (office use; Phase 1 is a web app, bills entered from paper)
Customers	3,000–5,000 now; must support 10,000 (NFR-02)
Daily volume	Several hundred bill entries per day (NFR-02)
Data type	Structured, finance-related (bottle balances, money owed, stock)
Documents	Printed bills, invoices, statements, POs, delivery/load sheets, customer quotations, factory dispatch notes, expense statements (generated, not uploaded)
Extra data from BRD v4.0	Agreed customer prices (a few rows per customer, with history), running expenses (tens of rows a month), customer quotations (a few a week) – negligible for sizing

Tier decision: TIER 1 (Small Business), with two small additions: off-server backups and an easy upgrade path. The staff count and concurrency match Tier 1. The data volume (10,000 customers, a few hundred bills per day) is small for PostgreSQL. Because the data is financial, backups are kept **off the server** as well (see 4.2).

2. Core Application Architecture (shared NUVI stack – unchanged)

Component	Recommendation
Frontend	Next.js + TypeScript
UI	Tailwind CSS + shadcn/ui (follows the approved prototype layout)
Backend	Spring Boot (Java 21), REST API
Database	PostgreSQL
ORM / Data access	Hibernate / Spring Data JPA
Authentication	JWT; passwords hashed with BCrypt; role-based access (Admin / Accountant / Delivery Staff) – NFR-05
PDF generation	OpenPDF on the backend – bills, monthly invoices, statements, POs, delivery & load sheets, customer quotations, dispatch notes, expense statements (FR-18, FR-23, FR-44, FR-52, FR-63, FR-65)

Component	Recommendation
Excel export	Apache POI – all reports (BRD section 8)
Email delivery	Not required for Phase 1 (see 5.1)
WhatsApp	WhatsApp Business Cloud API – Phase 1, Option A (see 5.2)
File storage	Local Docker volume (EBS) – only small files (scanned quotations, CSV imports)
Reverse proxy	Nginx
Containers	Docker Compose
CI/CD	GitHub Actions
Monitoring	Application logs + AWS CloudWatch basic alarm (CPU / disk)

System-specific design points

- **Audit and no-delete rule (BR-14, FR-57):** saved transactions are never deleted. Corrections are reversal entries, and every create/change is written to an audit table.
- **Numbering:** customer, product, supplier, PO, bill, invoice, dispatch, expense and quotation numbers are generated by the database sequences, never typed by users.
- **Pricing:** one Pricing Service gives the water price for a customer on a date (agreed price if one is in force, otherwise the standard price); the price is stored on each bill line so old bills never change.
- **Dates and currency:** stored as ISO dates / NUMERIC(12,2); shown as DD/MM/YYYY and LKR (NFR-07).
- **Future mobile app (NFR-08):** all business logic stays in the REST API, so a delivery-staff mobile app can be added later without changing the backend.

3. Deployment Flow & CI/CD

Step	Description
1. Developer laptop	Local development with Docker Compose (same layout as production)
2. GitHub repository	Branch + pull-request review before merge
3. GitHub Actions	Build, test and package Docker images on merge to the release branch
4. Shared UAT server	NUVI's shared UAT EC2 – client testing and sign-off
5. Production	Thuhina Water's own dedicated EC2 instance

Production layout: Internet → Nginx (HTTPS) → Next.js + Spring Boot → PostgreSQL – all in Docker Compose on one EC2 instance.

4. Hosting & Deployment (Tier 1)

4.1 Server

Component	Recommendation
Cloud / region	AWS – Asia Pacific (Mumbai) ap-south-1 (closest low-cost region to Sri Lanka)
Compute	EC2 t4g.small (2 vCPU / 2 GiB RAM) – Java heap capped at ~768 MB
OS	Ubuntu 24.04 LTS
Storage	30 GB gp3 EBS (OS + database + files)
Database	PostgreSQL in Docker on the same instance

Component	Recommendation
HTTPS	Nginx + Let's Encrypt (free SSL, auto-renew)
Domain	e.g. <code>app.thuhinawater.lk</code> – sub-domain of the client's own domain
Deployment model	One dedicated instance for Thuhina Water
Upgrade path	Resize to t4g.medium (4 GiB) if screens slow down as data grows – a few minutes' downtime, no redevelopment

4.2 Backup (NFR-06 – automatic daily backup, restore tested)

Item	Recommendation
Daily database dump	<code>pg_dump</code> every night at 11 pm, kept 7 days on the server
Off-server copy	Each dump copied to Amazon S3 (kept 30 days) – protects against losing the server
Server snapshot	Weekly EBS snapshot (kept 4 weeks)
Restore test	Restore the latest backup to the UAT server once a month and check totals

5. Communication Module

5.1 Email – not required for Phase 1

The BRD has no email requirement:

- Users are created by the Admin, and the Admin resets passwords. No self-service password reset by email is needed.
- Bills, invoices and statements are printed or saved as PDF (FR-52).
- Customer quotations (FR-65) are printed or saved as PDF; the **Email** button opens the user's own email program (`mailto:`) with the subject and message ready, and the user attaches the PDF. No server email is needed.

Recommendation: do not set up an email server. If the client later wants the system itself to **email quotations, monthly invoices or statements** to office/factory customers, add the standard NUVI option: the client's own Gmail / company email over SMTP with an App Password. It costs nothing and needs no separate server.

5.2 WhatsApp Business – included in Phase 1 (Option A – essential messages)

Client decision (Oct 2026): Option A only – "Left at door" notice, payment reminder and monthly invoice. Delivery reminders (Option B) are not in Phase 1 and can be switched on later.

This suits a water delivery business: most household customers use WhatsApp, and messages are cheaper and more reliable than printed reminders.

Component	Recommendation
Method	WhatsApp Business Cloud API (Meta) – official, no extra server needed
Account ownership	Thuhina Water owns the WhatsApp Business account and pays Meta directly
NUVI IT's role	Integrates the API only; does not own or bill for messages
Toggle	Enable / disable per message type in System Settings
Setup fields	Business phone number, Business Account ID, Access Token, Verify connection
Customer opt-in	Only customers who agree (WhatsApp number on the customer record)

Useful messages for this business

Message	When	Est. per month*
Delivery reminder	Evening before delivery (from Delivery Planning)	~11,000
"Left at door" notice	When a bill is saved as Left at door (FR-35)	~500
Bill / receipt	After a bill or payment is saved	optional
Payment reminder	Credit & Monthly bill customers with overdue balance	~400
Monthly invoice (PDF)	When monthly invoices are generated (FR-49)	~150

* Based on ~4,000 customers and ~450 deliveries per working day.

5.3 Estimated WhatsApp cost (if enabled)

Using the NUVI planning rate of **~US\$0.005 per utility message** (US\$1 ≈ Rs. 330):

Option	Messages / month	Est. cost / month
A – Essential: Left at door + payment reminders + invoices	~1,000–1,500	~US\$5–8 (~Rs. 1,650–2,650)
B – Full: Option A + delivery reminder for every delivery	~12,000–13,000	~US\$60–65 (~Rs. 19,800–21,500)

Meta charges per message, and the rate depends on the country and message type. **Check Meta's current rate card for Sri Lanka before quoting.** If the real utility rate is higher than the planning rate, Option B grows in proportion. Start with Option A and add delivery reminders after seeing real costs.

6. Additional Technical Notes

- **Keyboard bill entry (NFR-04):** the bill grid works fully from the keyboard, with server-side duplicate checks on paper bill numbers (BR-13).
- **Performance (NFR-02/03):** index customer name/phone/area, bill numbers and transaction dates. A few hundred bills per day stays well within Tier 1.
- **Data migration (FR-53–55):** a one-time CSV import before go-live (screen kept but hidden in the prototype for now).
- **Barcode / GS1:** not needed. Bottles are tracked by quantity and type, not individually (BR-15).
- **Out of scope (Phase 1):** mobile app, route/vehicle management, online ordering, payment gateway, automatic volume price bands, full accounting (only running expenses are recorded).

7. Cost Summary (per month, indicative)

Item	Est. cost / month
EC2 t4g.small	~US\$8–9
30 GB gp3 EBS	~US\$2–3
Public IPv4 address	~US\$3.6
S3 backups + weekly snapshots	~US\$1–2
Bandwidth	~US\$0 (well within AWS free outbound allowance)
Docker / Java / Spring / PostgreSQL / Nginx / Let's Encrypt	Free (open source)
Infrastructure total	~US\$15–18 (~Rs. 5,000–6,000)
Email (SMTP, if added later)	Free

Item	Est. cost / month
WhatsApp – Option A (Phase 1, paid by client to Meta)	~US\$5–8
WhatsApp – Option B (later, if enabled)	~US\$60–65
If upgraded to t4g.medium	add ~US\$8–9

Yearly items: domain renewal (client's existing domain, or a new .1k / .com – price as per registrar).

8. Assumptions & Notes

- All costs are planning estimates (AWS Mumbai region, US\$1 ≈ Rs. 330). Confirm with the AWS Pricing Calculator and Meta's rate card before any commercial quote.
- Tier 1 is chosen on users and concurrency. If Thuhina Water adds a delivery-staff mobile app (Phase 2) or many more office users, review the tier – moving tiers is a server resize, not a rebuild.
- WhatsApp message costs depend on business activity, not the server tier, and are paid by Thuhina Water directly to Meta.
- This document should be read with the BRD v4.0, the System Architecture & Design v1.1 and the approved prototype.